

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Pharm (Pharmacology and Toxicology)

University Theory Examination January 2018

PHCT003 Cellular and Molecular Basis of Pharmacology

Date: 09.01.2018, Tuesday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Write a note on apoptosis. 05
B) Write a note on antisense oligonucleotide therapy. 05
- Q.2 What are the functions of cellular adhesion molecules? Discuss the integrins as therapeutic target. 10

OR

- Q.2 A) Discuss the role of DNA cloning in receptor characterization. 10
B) Write a note on Polymerase chain reactions.
- Q.3 Answer the following (Any three) 15
(A) Discuss the applications of stem cell research.
(B) Write a note on selectins as therapeutic target.
(C) Discuss different steps involved in DNA replication.
(D) Write a note on excitatory amino acids and discuss their role in neuronal apoptosis.

SECTION-II

- Q.4 (A) Enlist various transport mechanism across cell membrane. Explain in detail facilitated and carrier mediated transport. 05
(B) Explain the importance of dose-response curve giving suitable examples. 05
- Q.5 (A) Classify purine receptors with emphasis on receptor type, mechanism of action and agonist-antagonist for the same. 05
(B) Discuss about Extracellular matrix (ECM). 05

OR

- Q.5 (A) Explain molecular structure of Cell membrane. 05
(B) Write a note GPIIb/IIIa receptor antagonists. 05
- Q.6 Answer the following (Any three) 15
(A) Discuss signal transduction mechanism by GPCR.
(B) Write a note on Cannabinoid receptors.
(C) Write a short note on aging and anti-aging neuromechanism.
(D) Write a note on different types of drug antagonism.